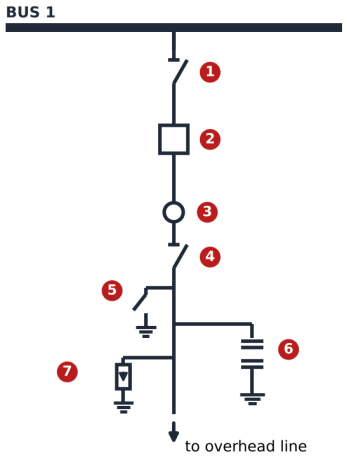
SUBSTATION EQUIPMENT

Made Easy

Circuit Breaker, Disconnector, Interlocking, CT, CVT, Surge Arrester, Transformer and the Single Line Diagram



1 Bus disconnector (DS)

2 Circuit breaker (CB)

3 Current transformer (CT)

4 Line disconnector (DS)

5 Earth switch (ES)

6 CVT (voltage for meters, relays, carrier)

7 Surge arrester (SA)

Read it top to bottom: BUS → isolate → break → measure → line. The CB is sandwiched between isolation points so it can be made safe for maintenance.

Built from your three voice notes, written for a beginner. Every topic is explained in plain words, with a diagram wherever a picture helps.

Study aid. Test limits, pass values and standard clause numbers must be checked against your project specification, the equipment manufacturer's manual and the current edition of the relevant IEC standard, and reviewed by a competent engineer on site.

What your voice notes asked for

Your three voice notes were one study plan for an interview or technical test on high-voltage substation equipment (this reading comes from the content; the notes do not say it outright). The speaker said these topics are what you will be asked, that there are 3 to 4 days left, and that you should read a little each day and memorise it. The table shows where each topic from the notes is covered.

Topic from the voice notes	Chapter
Single line diagram: learn to draw it, then explain each item in it	2
Circuit breaker: what it does, which tests at 220 kV and at 500 kV, how tests are performed	3
Disconnector (DS): what it does, how it differs from the breaker	4
Interlocking between the circuit breaker and the disconnector (and others)	5
CT: what it does	6
CVT: what it does	7
Surge arrester: what it does	8
Transformer: testing, vector group, hardware and connectors	9 and 10
How to study: use ChatGPT, read a little daily, memorise, plan for the last 3 to 4 days	11

How to use this book

Read chapter 1 first. It gives the whole picture in one page, so the later chapters make sense.

Each chapter ends with interview questions and short answers. Cover the answer and try to say it aloud.

Green boxes give an **in simple words** version. Amber boxes mark numbers and rules you must **verify** in your own project documents.

Contents

1. The Big Picture 4
2. The Single Line Diagram (SLD) 5
3. The Circuit Breaker (CB) 9
4. The Disconnector (DS) 14
5. Interlocking 16
6. The Current Transformer (CT) 19
7. The Capacitor Voltage Transformer (CVT) 21
8. The Surge Arrester 24
9. The Power Transformer 26
10. Hardware and Connectors 30
11. Your 4-Day Study Plan 31
Appendix A. Glossary 33

1. The Big Picture

A **substation** is a place where electricity is switched, measured, protected and changed from one voltage to another. Power travels at very high voltage because high voltage means low current, and low current means low loss over long distances.

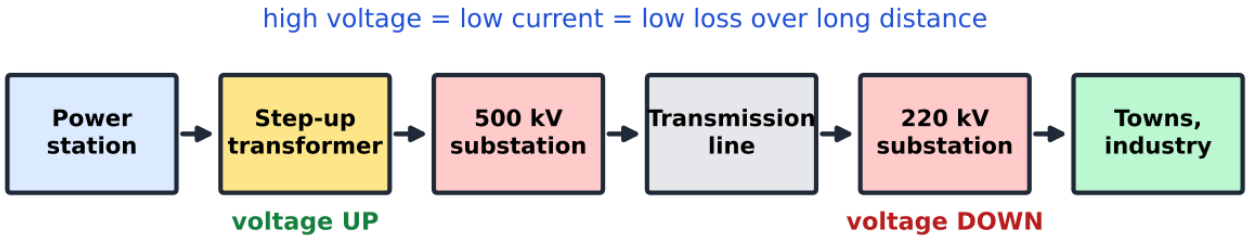


Figure 1.1 The journey of electric power. A substation sits at each change of voltage.

Think of the equipment as a team. Every item has exactly one main job.

Equipment	Short name	Its one job	Everyday comparison
Circuit breaker	CB	Switch the circuit ON and OFF, including automatically when a fault happens, and cut off very large fault current	A main switch that trips on overload, but far stronger
Disconnector (isolator)	DS	Give a visible, safe gap so people can work. It is NOT built to switch load current	Pulling the plug out of the wall, after the appliance is switched off
Earth switch	ES	Connect a dead circuit to earth so it cannot become live again	A safety strap
Current transformer	CT	Shrink a huge current to a small, safe copy for meters and relays	A translator for current
Capacitor voltage transformer	CVT	Shrink a huge voltage to a small, safe copy for meters and relays (also used for line carrier signals)	A translator for voltage
Surge arrester	SA	Send lightning and switching surges safely to earth	A pressure relief valve
Power transformer	TR	Raise or lower voltage	A gearbox for voltage

Voltage names

In everyday engineering talk, 220 kV is called **high voltage (HV)** and 500 kV is called **extra high voltage (EHV)**. The higher the voltage, the bigger and more carefully designed everything is, and the more checks it needs.

> **Remember**
> The order of equipment on a feeder: **bus, isolate, break, measure, line**.
> The disconnector isolates. The breaker interrupts. Never swap those two jobs in your mind.

2. The Single Line Diagram (SLD)

A **single line diagram** is a simple map of the electrical system. One line stands for all three phases, so the drawing stays clean. Your voice note says you will be asked to draw one and then explain each device in it. This chapter teaches exactly that.

The symbols

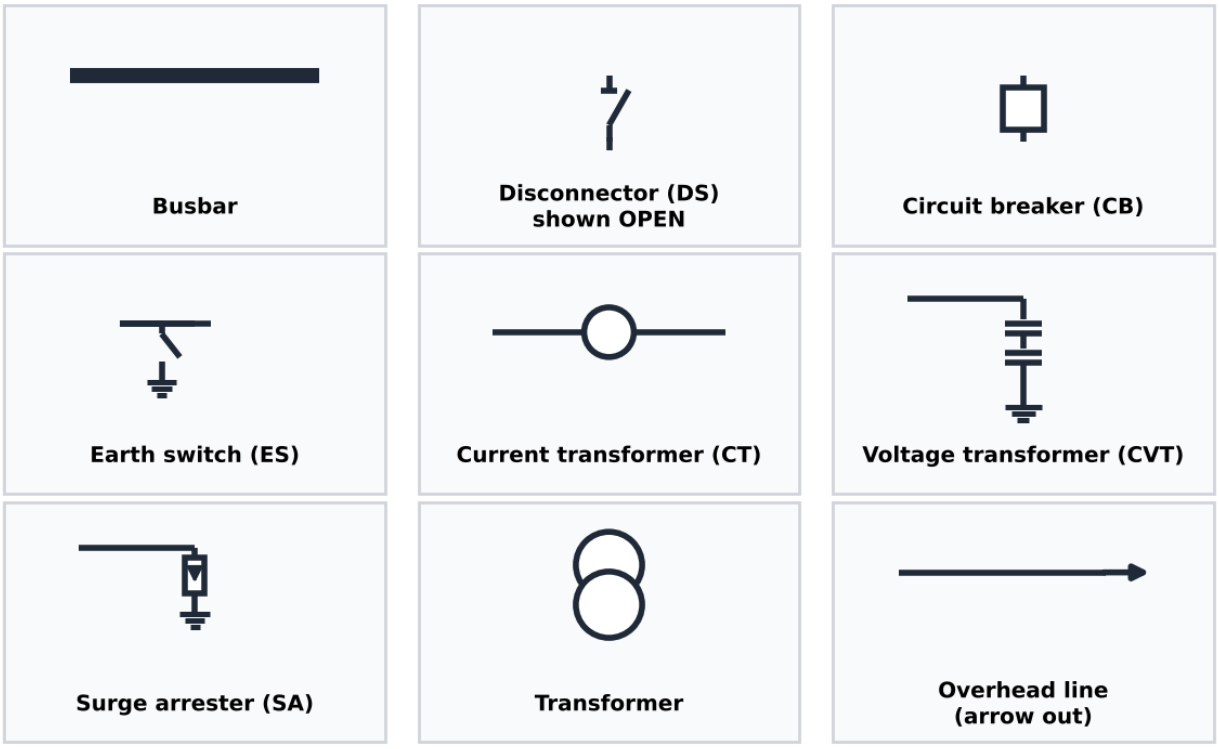


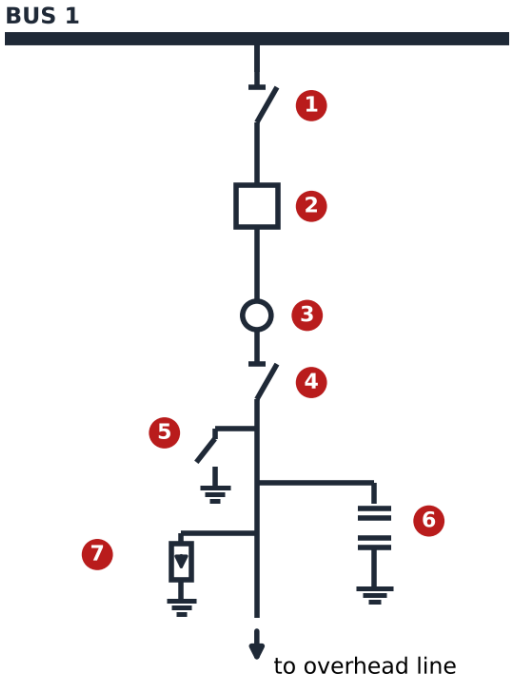
Figure 2.1 The nine symbols you need. Learn to draw each one in a few seconds.

In simple words

The breaker is a small **square**. The CT is a **circle on the line**. The disconnector is a **slanted blade**. The transformer is **two overlapping circles**. Anything that goes to earth ends in the three-line **ground** symbol.

A line bay, step by step

A **bay** is one complete circuit connection, from the busbar to the thing it feeds. This is the line bay for an overhead line.



1 Bus disconnector (DS)

2 Circuit breaker (CB)

3 Current transformer (CT)

4 Line disconnector (DS)

5 Earth switch (ES)

6 CVT (voltage for meters, relays, carrier)

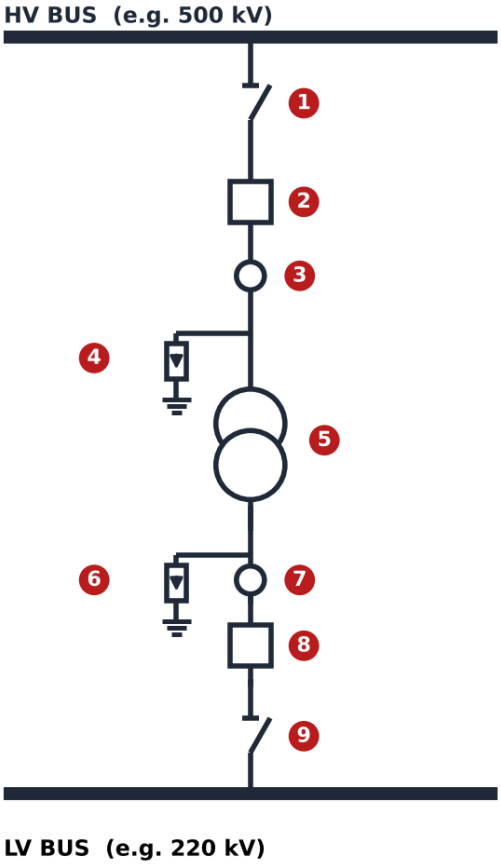
7 Surge arrester (SA)

Read it top to bottom: BUS → isolate → break → measure → line. The CB is sandwiched between isolation points so it can be made safe for maintenance.

Figure 2.2 A line bay. Follow the numbers from the top (busbar) to the bottom (line).

No.	Device	What it does here
1	Bus disconnector	Lets this bay be separated from the busbar for maintenance. Operated only when the breaker is open.
2	Circuit breaker	Switches the line on and off and clears faults automatically (when told by the protection relay).
3	CT	Reports the line current to the protection relay and the meter.
4	Line disconnector	Separates the breaker from the outgoing line, so the breaker can be made safe from both sides.
5	Earth switch	After isolation, earths the dead part so it cannot hold a dangerous charge.
6	CVT	Reports the line voltage to relays and meters. Can also couple carrier-frequency (PLC) signals onto the line, where used.
7	Surge arrester	Protects the equipment from lightning and switching surges arriving along the line.

A transformer bay



1 HV bus disconnector

2 HV circuit breaker

3 HV CT (protection of transformer)

4 Surge arrester, close to the HV terminals of the transformer

5 Transformer (shown as two coils; autotransformer at EHV is common)

6 Surge arrester on the LV side

7 LV CT

8 LV circuit breaker

9 LV bus disconnector

Arresters sit right next to the transformer because the transformer is the most expensive item and the most sensitive to voltage surges.

Figure 2.3 A transformer bay between a higher and a lower voltage busbar.

The layout repeats on both sides of the transformer: **disconnector, breaker, CT**. The surge arresters sit close to the transformer terminals.

Busbar arrangements (just the names)

- **Single busbar:** simplest and cheapest. If the bus is out, everything connected to it is out.
- **Double busbar:** each bay can connect to either of two buses, using two bus disconnectors. More flexible. Often used at 220 kV.
- **Breaker-and-a-half:** three breakers shared between two feeders. Very reliable. Often used at 500 kV.

> **Verify before you rely on it**
> 'Often used at 220 kV' and 'often used at 500 kV' are common practice, not a rule. Check the actual arrangement in your project's drawings.

How to draw an SLD in an exam, in 8 steps

1. Draw the **busbar** as a thick horizontal line and label it with its voltage.
2. Drop a vertical line from the bus for the bay.
3. Add the **bus disconnector** first.
4. Add the **circuit breaker** (square) below it.
5. Add the **CT** (circle) next to the breaker.
6. Add the **line disconnector** and, if asked, an **earth switch** on the line side.
7. Add the **CVT** and the **surge arrester** as branches going down to earth, on the line side.
8. Finish with the line arrow (or the transformer) and label every item with its name.

Interview questions

Q. Why is there a disconnector on both sides of the breaker?

A. So the breaker can be isolated from the live bus on one side and from the line on the other, and then earthed and worked on safely.

Q. Why is the CT drawn as a circle on the line?

A. Because the line passes through it. It is connected in series, so it sees the full line current.

Q. Where do the CVT and arrester connect?

A. Between the line and earth, on the line side of the line disconnector, so they protect and measure the incoming line.

Q. Which equipment is the single most expensive and most protected?

A. The power transformer, which is why arresters are placed right next to it.

3. The Circuit Breaker (CB)

What it does

The circuit breaker is the only device in the bay that can **safely interrupt fault current**. It carries load current all day, and when a fault happens it must open in a few cycles and put out the arc that forms between its contacts.

> **In simple words**
> A fault can push thousands of amps through the circuit. The breaker opens its contacts in a fraction of a second and stops the current. An arc tries to keep the current going, so the breaker uses a gas to cool and kill the arc.

How it puts out the arc

Alternating current passes through zero twice every cycle. A breaker is designed to cool and de-ionise the gap exactly around those current zero moments, so the current cannot start again. At 220 kV and 500 kV the gas used is **SF6** (sulphur hexafluoride), which is very good at quenching arcs and insulating. Lower-voltage breakers often use vacuum. Older breakers used oil or air.

Main parts

- **Interrupter (break):** the chamber where the contacts separate and the arc is quenched.
- **Insulators:** hold the live parts away from earth.
- **Operating mechanism:** the muscle that moves the contacts. Common types are spring, hydraulic, or a mix of both.
- **Control cabinet:** trip and close coils, auxiliary contacts, heaters, selector switches and wiring.
- **SF6 gas system:** density monitor with alarm and lock-out levels.

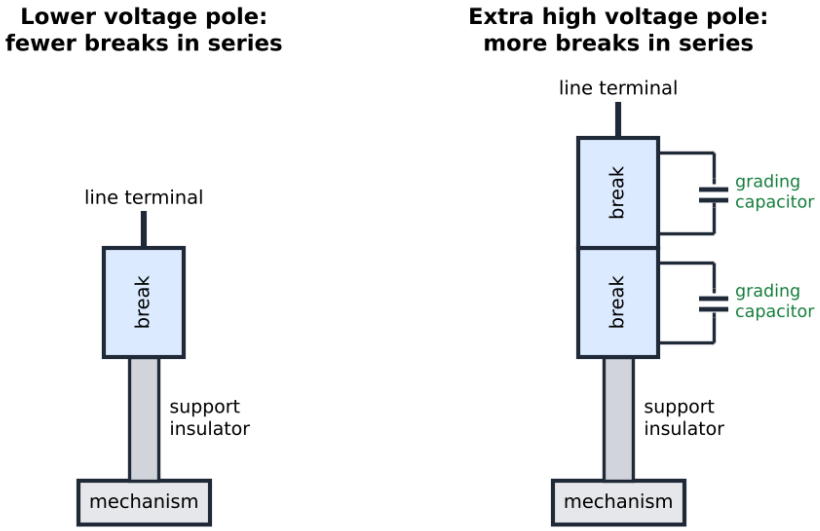


Illustration only. The real number of breaks depends on the manufacturer's design.

Figure 3.1 Higher-voltage breakers put more breaks in series, with grading capacitors to share the voltage between them.

Ratings you will see on the nameplate

- Rated voltage (kV) and rated normal current (A).
- Rated short-circuit breaking current (kA): the biggest fault the breaker can interrupt.
- Rated operating sequence: for example **O - 0.3 s - CO - 3 min - CO**, where O is open and CO is close-then-open. It describes how many fast reclosing duties the breaker can do.

Where the tests come from

Breakers are tested at three stages. **Type tests** prove the design. **Routine tests** are done on every breaker in the factory. **Site (commissioning) tests** prove it survived transport and was installed correctly. Your interview question is about the **site tests**, so those are listed below. The main standard for high-voltage AC breakers is **IEC 62271-100**.

Safety

Every test below is done on **isolated, earthed equipment** under a permit to work. Apply lock-out tag-out and keep a written record.

Abu Dhabi reference: ADOSH-SF Code of Practice **COP 15.0 Electrical Safety**, **COP 21.0 Permit to Work Systems**, **COP 24.0 Lock-out Tag-out (Isolation)** and **COP 2.0 Personal Protective Equipment**.

Remove test leads only after the equipment is discharged and re-earthed. Test sets can inject dangerous voltage or current.

Site tests on a circuit breaker

Test	What it checks	How it is done (short)	What you look for
Visual and mechanical check	Damage in transport, correct installation	Inspect nameplate, gas pressure gauge, bolts and torque marks, earthing, heaters, operations counter	No damage, correct readings, all fixings tight
Insulation resistance	Health of the insulation	Use a megger. Test the main circuit to earth with the breaker closed, and across the open contacts with the breaker open. Test control wiring separately at a lower voltage	High, steady values within the limits of the procedure
Contact resistance	Quality of the closed contacts	Breaker closed. Inject a DC test current with a micro-ohmmeter and measure the voltage drop across each pole	A low micro-ohm value, similar on all three poles, within the manufacturer's limit
Timing (operating time)	Speed and synchronism	Connect a breaker analyser. Operate close, open and close-open. Record each pole's time	Close time, open time and pole spread inside the manufacturer's table
Coil and control voltage	Control circuit health	Measure coil resistance. Find the lowest voltage at which the close and trip coils still operate the breaker	Operates at the low end of the allowed supply range
SF6 gas checks	Right gas, no leaks, dry gas	Check density monitor settings (alarm and lock-out), run a leak check, and measure moisture (dew point)	Values within the manufacturer's limits
Mechanism checks	Healthy energy storage	Measure spring charging time and motor current, or hydraulic pressure and pump run time	Within the manufacturer's range
Control logic checks	Protective functions work	Test anti-pumping, trip-free, pole discrepancy, lock-out on low gas, local/remote selector, and trip from the protection relay	Each function behaves exactly as designed
Power-frequency withstand (when specified)	Insulation under high voltage	Apply a high AC test voltage to the closed breaker to earth and across open contacts	No breakdown during the hold time

Verify before you rely on it

Check the following in the manufacturer's manual and the project test procedure: contact resistance limit, test current (often around 100 A DC or more), timing limits, coil operating voltage range (IEC commonly specifies roughly 85 to 110 % of rated voltage for the closing coil and 70 to 110 % for the opening coil), SF6 moisture and density limits, and the withstand test voltage.

How three important tests are performed

Contact resistance

1. Make sure the breaker is isolated, earthed, and then connected only to the test set (earths removed from the measuring points under the permit).
2. Close the breaker.
3. Connect the micro-ohmmeter current leads outside the voltage leads, on each side of the pole.
4. Inject the DC test current and read the resistance in micro-ohms.

5 Record all three poles. Compare with the manufacturer's limit and the factory value. Large or uneven values mean a poor contact.

Timing test

1 Connect the analyser to the main contacts of each pole and to the close and trip coil circuits.

2 Give a close command and record the time from command to contact touching.

3 Give an open command and record the time from command to contact separation.

4 Do a close-open operation and record the pole spread, the gap between the first and the last pole.

5 Compare each value with the manufacturer's table.

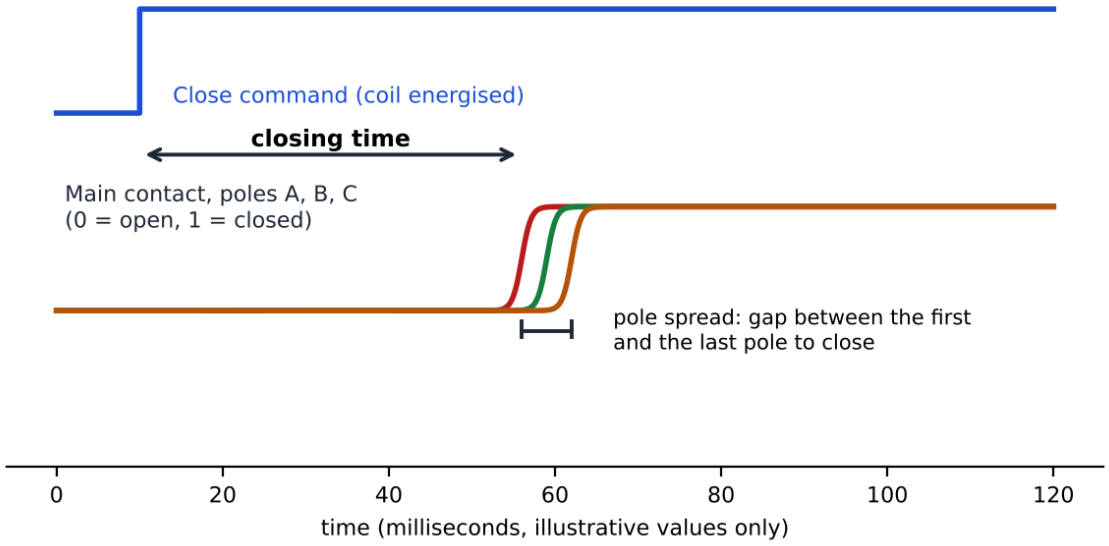


Figure 3.2 What a timing record shows. Real values come from the manufacturer's manual.

Insulation resistance

1 Check the area is isolated, earthed and the permit is in place.

2 Disconnect the earths on the points to be tested, one at a time.

3 Choose the megger voltage required by the procedure and apply it for the specified time.

4 Read and record the value.

5 Discharge the circuit, then restore the earth.

What changes between 220 kV and 500 kV?

The honest answer: the **list of test types is mostly the same**. What changes is how many places you must test and a few extra items that only exist at the higher voltage.

Point	220 kV breaker	500 kV breaker
Test family	Insulation, contact resistance, timing, coils, gas, mechanism, control logic	The same family

Point	220 kV breaker	500 kV breaker
Breaks per pole	Fewer breaks in series	More breaks in series, so contact resistance and timing may be checked per break as the manufacturer specifies
Grading capacitors	Present only on some designs	Often fitted across each break. Test capacitance and tan delta (loss factor) of each capacitor
Pre-insertion resistors (PIR)	Not usual	Fitted on some designs to limit switching surges on long lines. Test the resistor value and its insertion time
Controlled switching	Rare	Used on some projects. Check the controller's settings and the timing it relies on
Withstand test voltage	Lower	Much higher. Take the figure from the standard and the project procedure
Gas volume and size	Smaller	Larger. Gas handling, leak checks and moisture checks take longer
Independent pole operation	Common, to allow single-phase auto-reclose	Common. Timing and pole discrepancy checks are important

Verify before you rely on it

Which of these extra items apply depends on the manufacturer and the project specification. In an interview, say: 'The test list is the same family. At 500 kV there are more breaks, grading capacitors and sometimes pre-insertion resistors, so there are extra checks for those.'

Interview questions

Q. What is the main job of a circuit breaker?

A. To make and break the circuit under normal load and to interrupt fault current safely, automatically on a command from the protection relay.

Q. Which gas is used at 220 kV and 500 kV, and why?

A. SF6, because it quenches the arc well and is an excellent insulator.

Q. What does the contact resistance test tell you?

A. How good the closed contacts are. A high or uneven value means poor contact and local heating.

Q. What is pole spread?

A. The time difference between the first and the last pole to operate. Too much spread causes unbalanced currents and stress.

Q. What is SF6 moisture checked for?

A. Water in the gas can form acids when arced and weakens the insulation, so it must stay within the limit.

Q. Which standard covers high-voltage circuit breakers?

A. IEC 62271-100 for AC circuit breakers, with common requirements in IEC 62271-1.

4. The Disconnector (DS)

What it does

A disconnector (also called an **isolator**) opens a circuit **when there is no current**, to create a **visible, safe gap**. It cannot put out an arc. If you open it with load current flowing, a dangerous arc forms and can destroy the equipment and hurt people.

> **Golden rule**
> A disconnector **never makes or breaks load current**. Only the circuit breaker does that. Always open the breaker first and close it last.

The standard allows a disconnector to switch only very small currents, such as the charging current of a short piece of bus or the small current of a transformer's no-load magnetisation. The relevant standard is **IEC 62271-102**.

Disconnector versus circuit breaker

Question	Disconnector	Circuit breaker
Can it open under load current?	No	Yes
Can it interrupt a fault?	No	Yes
Gives a visible gap?	Yes (most types)	No, the contacts are inside a sealed chamber
Operation speed	Slow, motor driven	Very fast
Main purpose	Isolation for safety	Switching and protection

Types you may be asked about

- **Centre-break (horizontal):** two arms rotate and meet in the middle.
- **Double-break:** two gaps in series for higher voltage.
- **Vertical-break:** the blade lifts vertically.
- **Pantograph:** a folding arm that reaches up to the overhead bus. Saves space.

Earth switch (ES)

The earth switch is the safety partner of the disconnector. After you open the disconnectors, you close the earth switch to connect the dead circuit to earth. Earth switches are often built into the disconnector and are **mechanically interlocked** with it, so the main blade and the earth blade can never both be closed.

Site tests on a disconnector

Test	What you check
Visual and alignment	Blades and contacts line up and close fully, insulators are clean and undamaged
Contact resistance	Low, even resistance of the main contacts when closed
Insulation resistance	Insulation of the support insulators and of the control wiring
Operating test	Smooth motor operation, correct operating time and motor current, manual handle works, correct end position and limit switch settings
Interlock test	Operation is blocked when the interlock conditions are not met (see chapter 5)
Earth switch test	Earth blade makes proper contact and cannot close while the main blade is closed

Interview questions

Q. Can a disconnector be opened under load?

A. No. It has no arc-quenching ability. Open the breaker first.

Q. Why do we use a disconnector at all if the breaker already opens the circuit?

A. The breaker's gap is inside a sealed chamber and cannot be seen. The disconnector gives a visible isolation gap that workers can trust.

Q. What is an earth switch for?

A. To connect a dead circuit to earth so it cannot become live by accident or hold a charge.

5. Interlocking

Why interlocks exist

Interlocking stops people from operating equipment in the wrong order. It protects people first and equipment second. Your voice note asks about the interlocking of the circuit breaker and the disconnector, and who else the breaker is interlocked with.

The simple logic

Operation	Allowed only when
Open or close a bus or line disconnector	The circuit breaker of that bay is OPEN, and the related earth switch is OPEN
Close an earth switch	Every disconnector that can supply that part is OPEN (and ideally the part is proved dead)
Close a disconnector	The earth switch on that part is OPEN
Close the circuit breaker	The disconnectors are fully in position (not half-way), the earth switch is open, and the breaker's own conditions are healthy (gas, energy stored, no lock-out)

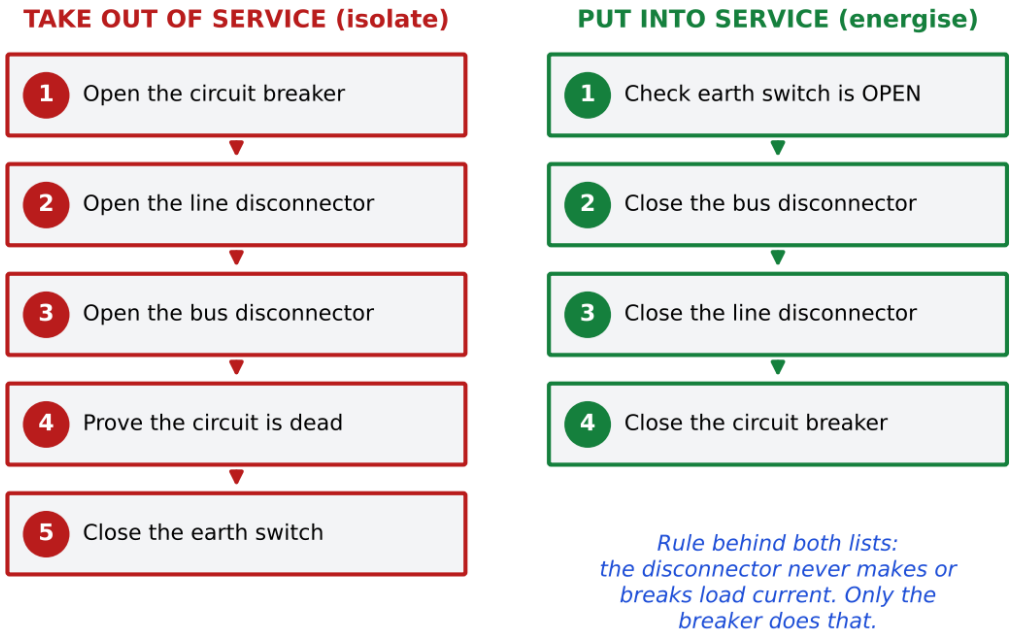


Figure 5.1 The safe order of operations. The interlocks make it impossible to go out of order.

Who is the breaker interlocked with?

- The **disconnectors** on both sides (the main interlock).
- The **earth switches** on the same section.
- The **protection lock-out relay (device 86)**, which blocks closing after a protection trip until someone resets it.
- The breaker's **own conditions**: low SF6 gas, spring not charged, low hydraulic pressure, or a pole discrepancy.
- The **local/remote selector**, so only one place has control at a time.

Kinds of interlock

Kind	How it works
Mechanical (key)	Physical keys or linkages. A key is trapped until the earlier step is done
Electrical	Auxiliary contacts of breakers and switches wired into the control circuit
Software (bay controller)	Interlock logic inside the substation control system. It checks equipment positions before it allows a command
Bypass key	An emergency override. Use it only with written authority and record it

Bus coupler exception

On a double busbar, a bay can move from one bus to the other while it stays live. A bus disconnector may then be allowed to operate with the bus coupler breaker closed, because both buses are at the same voltage and no load current is switched by the disconnector. This is a special condition in the interlock design.

> **Verify before you rely on it**
> The exact interlock table is set by the project (the 'interlocking schedule'). Use the logic above for the interview and the project document for real work.

How interlocks are tested

At commissioning, the engineer tries every forbidden operation and checks it is **blocked**, then tries the allowed operation and checks it works. Typical examples: try to close a disconnector while its breaker is closed, and try to close an earth switch while a disconnector is closed. Both must be refused.

Interview questions

Q. What is the interlock between the breaker and the disconnector?

A. The disconnector can be operated only when its breaker is open. This stops the disconnector making or breaking load current.

Q. In which order do you isolate a feeder?

A. Open the breaker, open the line disconnector, open the bus disconnector, prove dead, then close the earth switch.

Q. In which order do you energise it?

A. Check earths are open, close the bus disconnector, close the line disconnector, then close the breaker.

Q. What is device 86?

A. A lock-out relay. After a protection trip it keeps the breaker from closing until it is reset deliberately.

6. The Current Transformer (CT)

What it does

A CT turns a very large primary current into a small secondary current that is an accurate copy, usually **1 A or 5 A**. Meters and protection relays are fed from this small current, so they stay small, cheap and safe.

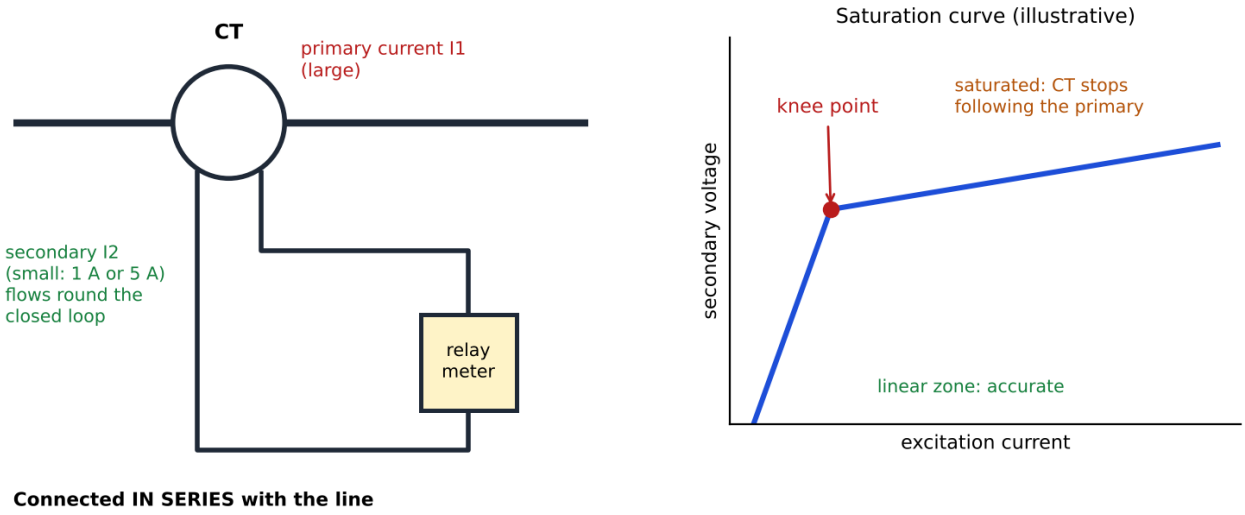


Figure 6.1 A CT is connected in series with the line. Its secondary feeds relays and meters. Right: the saturation curve.

A worked example

> **Example**
>
> A CT has a ratio of **1200/1 A**. The line carries **600 A**.
>
> Secondary current = 600 x (1 / 1200) = **0.5 A**. The relay sees 0.5 A and knows the line carries 600 A.

Metering CT and protection CT

Type	Purpose	Requirement	Example class
Metering	Billing and measurement	Very accurate at normal current, and it is happy to saturate during a fault to protect the meter	0.2, 0.2S, 0.5
Protection	Feed relays during faults	Must stay accurate at many times the rated current so the relay sees the real fault current	5P20 (error within 5 % at 20 times rated current)

Many CTs have **several cores**, some for metering and some for protection, in one housing. Higher-voltage CTs can have four or more cores.

Burden and saturation

- **Burden** is the load on the secondary (relay, meter and wire), measured in VA or ohms. Too much burden makes the CT inaccurate.
- **Saturation** happens when the core cannot carry more flux. The output stops following the input (right side of Figure 6.1). Protection CTs are chosen with a high enough **knee-point voltage** so they do not saturate during the faults the relay must see.

> **Danger**
> **Never open the secondary circuit of a CT while the primary carries current.** With no secondary current to balance it, the core is driven far into saturation and a very high voltage (can reach thousands of volts) appears across the open terminals. It can kill a person, burn the CT and cause a fire.
>
> Always short the secondary terminals first, then disconnect the burden. Test blocks have shorting links for this.

Site tests on a CT

Test	What it shows
Ratio test	The real ratio matches the nameplate on every tap and core
Polarity test	P1 and S1 are the same polarity. Wrong polarity makes protection misoperate
Insulation resistance	Insulation of the primary and secondary windings
Secondary winding resistance	Health of the winding, and input to burden calculations
Excitation (saturation) curve	The knee-point voltage, which proves the CT will not saturate too early
Burden measurement	The connected load is within the CT's rating
Capacitance and tan delta (higher-voltage, oil-paper types)	Ageing or moisture in the main insulation

Interview questions

Q. What does a CT do?

A. It steps down line current to a small standard value (1 A or 5 A) for meters and relays.

Q. Is a CT connected in series or in parallel?

A. In series, so the full line current flows through its primary.

Q. Why must a CT secondary never be left open?

A. Because a dangerously high voltage develops across the open secondary.

Q. What does 5P20 mean?

A. A protection class. The error is within 5 % at 20 times the rated current.

Q. What is the knee-point voltage?

A. The secondary voltage above which the CT starts to saturate.

7. The Capacitor Voltage Transformer (CVT)

What it does

A CVT steps a very high line voltage down to a small, accurate secondary voltage, commonly **110/√3 V** (about 63.5 V) per phase, for meters, relays and synchronising. At very high voltage it is cheaper and more practical than a conventional wound voltage transformer.

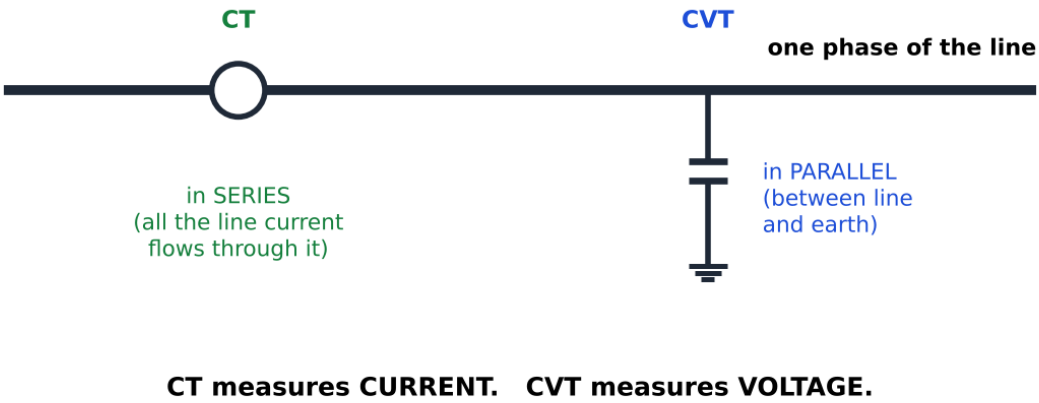


Figure 7.1 CT and CVT compared. The CT is in series. The CVT is across the line and earth.

How it works, in two stages

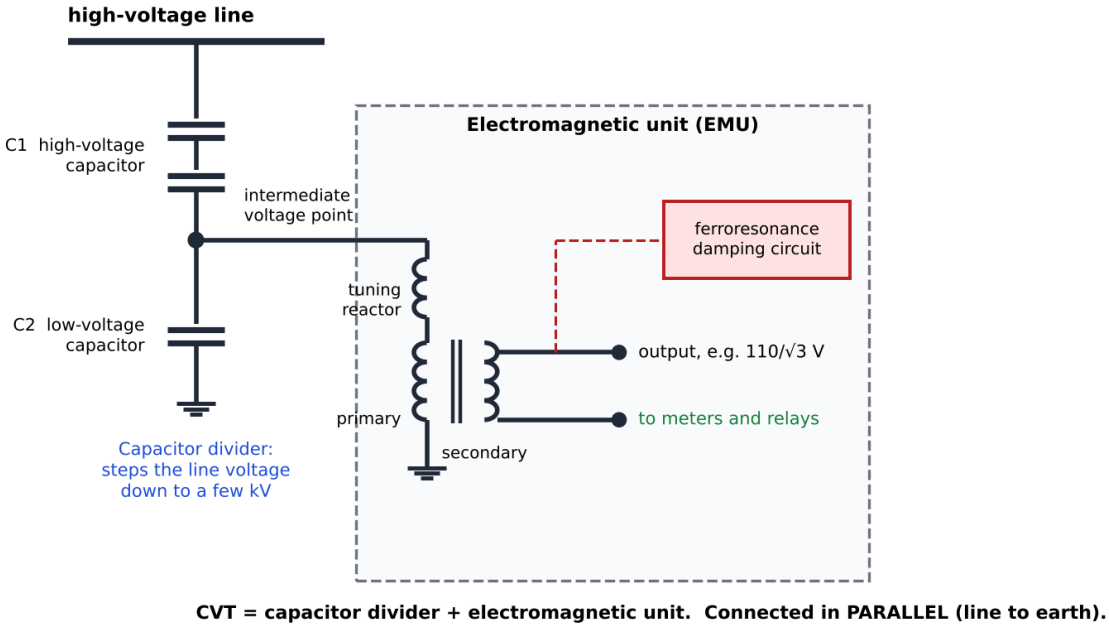


Figure 7.2 Inside a CVT: a capacitor divider, then an electromagnetic unit.

1. **Capacitor divider (C1 and C2):** two capacitor stacks in series between the line and earth. The voltage divides between them, and the small share across C2 (a few kV) is taken out as an intermediate voltage.
2. **Electromagnetic unit (EMU):** a transformer steps the intermediate voltage down to the standard secondary value. A **tuning reactor** cancels the capacitor's reactance at power frequency, which keeps the output accurate.
3. A **ferroresonance damping circuit** stops a harmful oscillation between the transformer core and the capacitors.

The capacitor stack can also couple **power-line carrier (PLC)** communication signals onto the line, which is a second use at some feeders.

CT versus CVT

Point	CT	CVT
Measures	Current	Voltage
Connection	In series with the line	In parallel, between line and earth
Secondary	1 A or 5 A	About 110/√3 V or similar
Never do this	Never open the secondary circuit	Never short the secondary circuit
Extra use	None	Carrier (PLC) coupling

> **Remember**
> The opposite warnings: a CT secondary must not be **open**. A CVT or voltage transformer secondary must not be **short-circuited**.

Site tests on a CVT

Test	What it shows
Ratio and polarity	Correct step-down ratio and winding polarity
Capacitance and tan delta of C1 and C2	Ageing or moisture in the capacitors. The most important test of the capacitor stack
Insulation resistance	Insulation of the capacitor stack, the EMU and the secondary wiring
Burden and accuracy check	The connected load is within the CVT's rating and the output is in its accuracy class
Damping circuit check	The ferroresonance damping circuit is intact and correctly connected

Verify before you rely on it

The secondary value 110/√3 V is typical. Confirm the exact rated secondary and accuracy class on the nameplate.

Interview questions

Q. What does a CVT do?

A. It steps the line voltage down to a small standard voltage for meters, relays and synchronising.

Q. Why use a CVT instead of an ordinary voltage transformer at 220 kV and 500 kV?

A. A capacitor divider is cheaper and more compact than a very large wound transformer at these voltages. It can also carry PLC signals.

Q. Why is there a tuning reactor?

A. It cancels the capacitor reactance at power frequency so the output voltage stays accurate.

Q. What is ferroresonance?

A. An unstable oscillation between a nonlinear transformer core and the capacitors. The damping circuit stops it.

8. The Surge Arrester

What it does

A surge arrester protects equipment from **overvoltage surges**. At normal voltage it carries almost no current. When a surge arrives it turns into a good conductor and sends the surge current to earth. When the surge is gone it goes back to being an insulator.

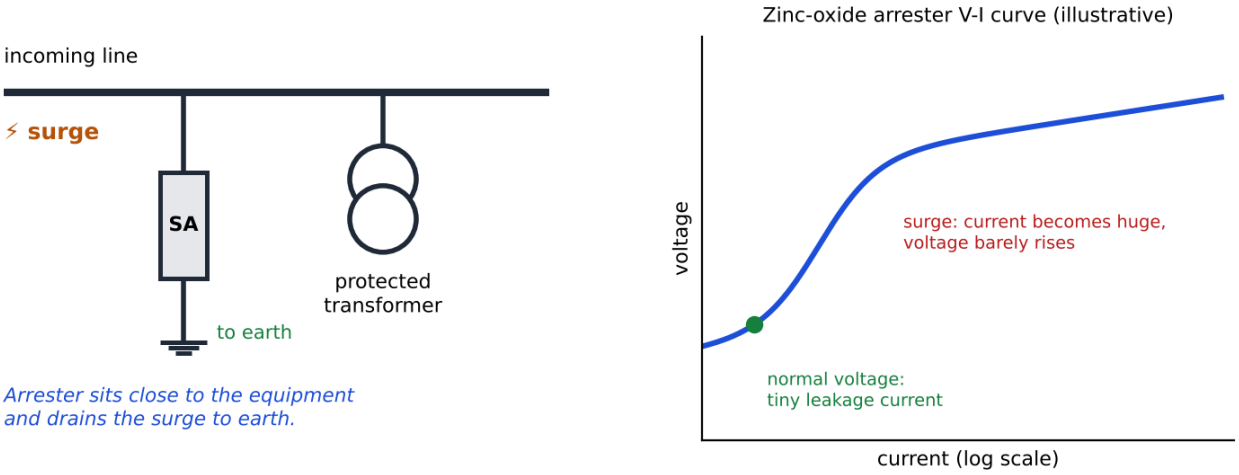


Figure 8.1 Left: an arrester next to a transformer. Right: why it works, a very non-linear V-I curve.

Where surges come from

- **Lightning:** very fast, very high.
- **Switching:** caused by operating breakers and disconnectors, slower but still dangerous.
- **Temporary overvoltage:** lasts longer, for example after a fault or load rejection.

Construction

Modern arresters at these voltages are **metal-oxide (zinc-oxide)** without spark gaps. A stack of zinc-oxide blocks sits inside a porcelain or polymer housing. They are rated by **continuous operating voltage** and **rated voltage**, and by a nominal discharge current class. The relevant standard is **IEC 60099-4**.

Placement

- At the **line entry** of the substation.
- **Close to the transformer terminals** (the closer, the better the protection).
- A **surge counter** and a **leakage current monitor** are often fitted on the earth lead.

Site tests on an arrester

Test	What it shows
Insulation resistance	Basic condition of the unit
Leakage current (total and resistive)	A rising resistive leakage means the zinc-oxide blocks are ageing
Surge counter check	The counter and its leakage meter work
Earthing connection	A low-resistance, sound path to earth
Thermography (infrared)	Hot spots on a running arrester

Interview questions

Q. What does a surge arrester do?

A. It diverts lightning and switching surges to earth so that the equipment does not see a damaging voltage.

Q. Why is it placed close to the transformer?

A. The surge voltage rises along the connecting leads and travelling waves reflect, so a long distance reduces the protection.

Q. What material is used?

A. Zinc oxide (metal oxide) blocks without spark gaps.

Q. How do you know an arrester is ageing?

A. Its resistive leakage current rises compared with earlier readings.

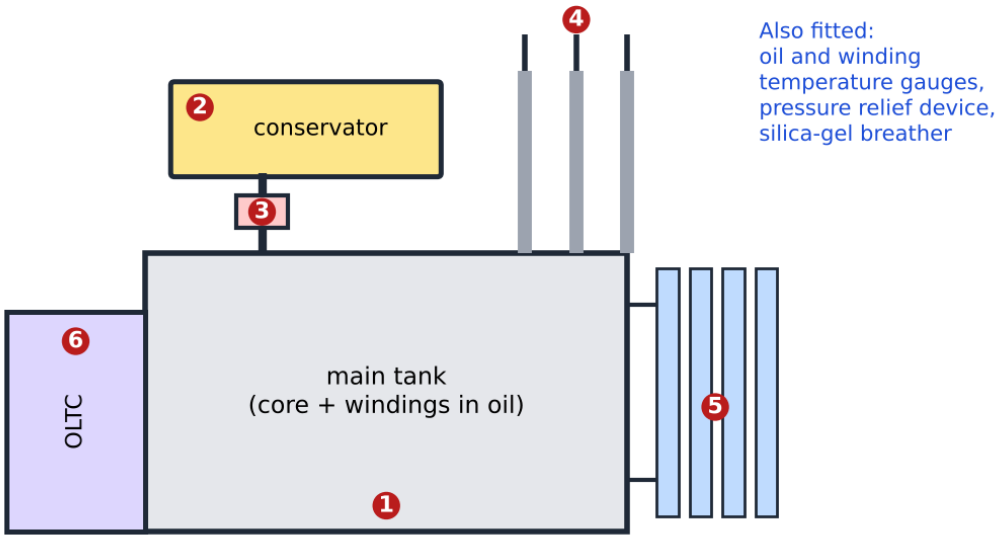
9. The Power Transformer

What it does

A transformer changes voltage. The voltage ratio equals the turns ratio: **V1 / V2 = N1 / N2**. A step-up transformer raises voltage for transmission. A step-down transformer lowers it for distribution. For a 500/220 kV unit the ratio is about 500 / 220, roughly 2.3 to 1.

Raising the voltage lowers the current for the same power. With half the current, the I^2R loss in the line is one quarter. That is why transmission uses high voltage.

Main parts



1 Tank: holds the core, windings and insulating oil
2 Conservator: oil expansion tank on top
3 Buchholz relay: gas/oil surge relay in the pipe between tank and conservator
4 Bushings: take the line terminals through the tank safely
5 Radiators/coolers: remove heat (ONAN, ONAF, OFAF...)
6 On-load tap changer (OLTC): adjusts voltage ratio under load

Figure 9.1 The main parts of an oil-filled power transformer.

- **Core and windings:** the active part, made of laminated steel and copper (or aluminium) coils.
- **Oil:** insulates and carries heat to the radiators.
- **Bushings:** insulated entry points for the terminals.
- **Cooling:** coded as ONAN (oil natural, air natural), ONAF (fans), OFAF (pumps and fans), and so on.

At the 500/220 kV level the interconnecting transformers are commonly **autotransformers**, where part of the winding is shared between the high and low sides. This is smaller and cheaper than a two-winding unit.

Protective devices on the transformer

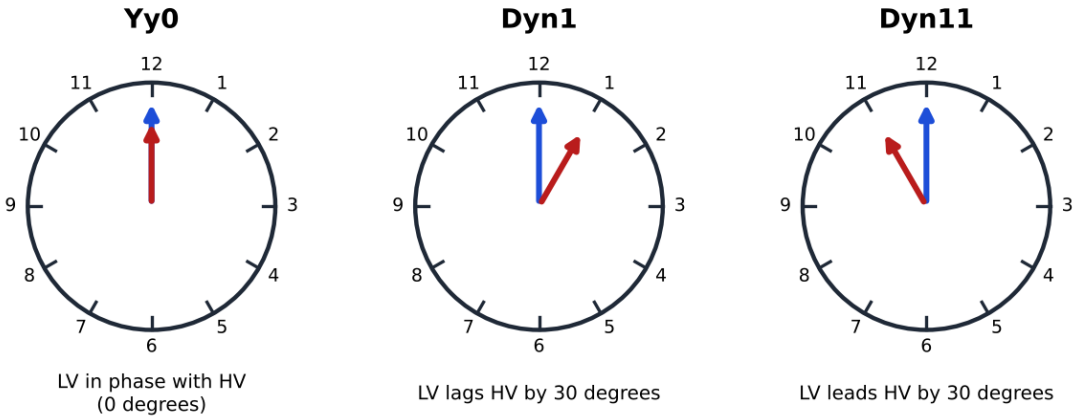
Device	Job
Buchholz relay	Detects gas or a sudden oil surge caused by an internal fault. Gives alarm, then trip
Pressure relief device	Opens to release dangerous internal pressure
Oil and winding temperature indicators	Alarm and trip on overheating, and control the cooling fans
Oil level indicator	Warns of low oil
Silica-gel breather	Dries the air that enters when the oil breathes

The vector group

Three-phase transformer windings can be connected in star (**Y**), delta (**D**) or zigzag (**Z**). The **vector group** is a short code that tells you how the high-voltage and low-voltage windings are connected, and the **phase shift** between them.

How to read the code

- **Capital letters** are for the high-voltage winding. **Lower-case letters** are for the low-voltage winding.
- **N or n** means the neutral is brought out for earthing.
- The **number** is the **clock position** of the LV phase voltage when the HV phase voltage is placed at 12 o'clock. Each hour is **30 degrees**.



Blue arrow = HV phase A, always at 12 o'clock. Red arrow = LV phase a, the hour in the name.

Figure 9.2 Three vector groups as clock faces.

Worked example: Dyn11

D: HV winding in delta. **y**: LV winding in star. **n**: LV neutral brought out.

11: the LV voltage is at the 11 o'clock position, which means the LV voltage **leads** the HV voltage by 30 degrees (11 hours is 330 degrees lagging, or 30 degrees leading).

Group	Meaning	Phase shift (LV relative to HV)
Yy0	Star-star, in phase	0 degrees
Dyn1	Delta-star with LV neutral, 1 o'clock	LV lags HV by 30 degrees
Dyn11	Delta-star with LV neutral, 11 o'clock	LV leads HV by 30 degrees
YNa0d1 (typical EHV autotransformer with tertiary)	Auto-connected star HV and LV, in phase, with a delta tertiary at 1 o'clock	LV in phase with HV; the tertiary lags by 30 degrees

Why the vector group matters

- **Parallel operation:** two transformers feeding the same bus must have the **same vector group**, otherwise a large circulating current flows.
- **Phase shift:** it must match the network connection and the protection settings.
- **Earthing and harmonics:** a delta winding traps the third-harmonic currents and gives a path for earth faults.

Site tests on a power transformer

Test	What it shows
Visual, oil level and accessories	No damage in transport. Correct oil level. All accessories fitted
Insulation resistance and polarisation index	Dryness and condition of the insulation. Polarisation index is the ratio of the 10-minute to the 1-minute reading
Winding resistance	Good joints and tap changer contacts, even across phases
Turns ratio (TTR) on all tap positions	Correct voltage ratio. The test also proves the vector group
Vector group (phase displacement) test	That the connection matches the nameplate
Excitation (no-load) current	Healthy core and winding
Capacitance and tan delta (windings and bushings)	Moisture or ageing of the insulation and bushings
Sweep frequency response analysis (SFRA)	Mechanical condition of the windings, compared with a baseline
Oil tests: breakdown voltage, moisture, dissolved gas analysis (DGA)	Condition of the oil and early signs of internal faults
Protection and auxiliary checks	Buchholz, pressure relief device, temperature trips, fans, pumps and tap changer operate correctly

Interview questions

Q. What is a vector group?

A. A code for how the HV and LV windings are connected and the phase shift between them, read as a clock position.

Q. What does Dyn11 mean?

A. HV delta, LV star with neutral, LV voltage leading HV by 30 degrees.

Q. Why must parallel transformers have the same vector group?

A. A different phase shift causes a large circulating current between them.

Q. What does the Buchholz relay detect?

A. Gas or a sudden oil surge from an internal fault.

Q. Why are 500/220 kV transformers often autotransformers?

A. Because they are smaller, lighter and cheaper for a given rating when the voltage ratio is not very large.

10. Hardware and Connectors

Your voice note mentions hardware and connectors only briefly, so this is a short overview. They join the big equipment together and carry the current between them.

Item	What it is for
Conductor (for example ACSR or aluminium alloy) and tubular aluminium bus	Carries the current between equipment
Clamps and connectors (T-clamps, parallel groove, terminal connectors)	Join conductors to each other and to equipment terminals
Bimetallic connectors	Join aluminium to copper without galvanic corrosion
Expansion connectors	Let the bus expand and contract with temperature
Corona rings and shields	Smooth the electric field at EHV so that corona discharge and radio noise are reduced
Spacers (for bundled conductors)	Keep the conductors of a bundle apart at EHV
Insulator strings	Support and insulate the conductor from the steel structure

- **Why bimetallic?** Aluminium and copper in direct contact corrode over time, and the joint heats up. A bimetallic piece separates them.
- **Checks:** torque on every bolt, contact resistance of important joints, and infrared thermography on energised joints. A hot joint is a failing joint.

11. Your 4-Day Study Plan

Your voice note says: read a little each day, memorise, and use ChatGPT as a coach. Here is a simple plan for the last 3 to 4 days.

Day	Study	Practise
1	Chapter 1 (big picture) and chapter 2 (SLD)	Draw a line bay and a transformer bay from memory, 3 times each
2	Chapters 3, 4 and 5: breaker, disconnector, interlocking	Say the isolate and energise sequences aloud. List the CB site tests without looking
3	Chapters 6, 7 and 8: CT, CVT, surge arrester	Explain CT versus CVT in 30 seconds. Explain why the CT secondary must never be open
4	Chapters 9 and 10: transformer, vector group, hardware	Work out 3 vector groups on a clock. Then run a full mock interview with the question list below

Prompts to give ChatGPT (copy and paste)

- "List the site commissioning tests for a 220 kV SF6 circuit breaker and explain how each is performed, in simple words."
- "What extra tests apply to a 500 kV circuit breaker compared with a 220 kV one, such as grading capacitors and pre-insertion resistors?"
- "Explain the interlocking between a circuit breaker, bus disconnector, line disconnector and earth switch in a double busbar substation."
- "Describe how to draw a single line diagram for a 220 kV line bay and explain each device."
- "Explain the difference between a CT and a CVT and what each does in the bay."
- "What does a surge arrester do and where is it placed?"
- "Explain transformer vector groups with Dyn11 and YNd1 using a clock diagram."
- "Ask me 20 interview questions on substation equipment, one at a time, and correct my answers."

> **Remember**
> Ask the AI to name the standard (for example IEC 62271-100) next to each answer, then confirm the clause yourself. AI answers can sound sure and still be wrong on numbers.

Rapid-fire questions for your last revision

Q. Which device interrupts fault current?

A. The circuit breaker.

Q. Which device gives a visible isolation gap?

A. The disconnector.

Q. Which device earths a dead circuit?

A. The earth switch.

Q. Which device feeds relays with a small current?

A. The CT.

Q. Which device feeds relays with a small voltage?

A. The CVT (or a voltage transformer).

Q. Which device protects against lightning and switching surges?

A. The surge arrester.

Q. Which device changes the voltage level?

A. The power transformer.

Q. What is the first step to take a feeder out of service?

A. Open the circuit breaker.

Q. What is the last step to make it safe?

A. Close the earth switch after proving the circuit is dead.

Q. Which is never opened with its secondary open?

A. A CT.

Q. Which standard covers disconnectors and earth switches?

A. IEC 62271-102.

Q. Which standard covers metal-oxide surge arresters?

A. IEC 60099-4.

Q. Which standard family covers instrument transformers?

A. IEC 61869 (part 2 for CTs, part 5 for CVTs).

Q. Which standard family covers power transformers?

A. IEC 60076.

Final note

This book is a learning aid made from your voice notes and general engineering knowledge. It is not a design document or a test procedure. Test values, limits and procedures must be taken from your project specification and the equipment manufacturer's manual and checked by a competent engineer on site.

Appendix A. Glossary

Term	Meaning
Bay	One complete circuit connection from the busbar to the equipment it feeds
Burden	The load on a CT or VT secondary, in VA or ohms
CB	Circuit breaker
CT	Current transformer
CVT	Capacitor voltage transformer
DS	Disconnector, also called isolator
ES	Earth switch
EHV	Extra high voltage, for example 500 kV
HV	High voltage, for example 220 kV
Interlock	A rule or device that blocks an unsafe operation
Knee-point voltage	The voltage above which a CT starts to saturate
OLTC	On-load tap changer, adjusts the transformer ratio while loaded
PIR	Pre-insertion resistor in some EHV breakers
PLC	Power line carrier, communication over the power line
Pole spread	Time difference between the first and last pole to operate
SA	Surge arrester
SF6	Sulphur hexafluoride gas used for insulation and arc quenching
SLD	Single line diagram
Tan delta	Dissipation factor, a measure of insulation loss
Vector group	Code for winding connections and phase shift of a transformer

Appendix B. Items to verify before you rely on them

- Contact resistance limits and the test current for the specific breaker (manufacturer's manual).
- Timing limits and pole spread limits for the specific breaker (manufacturer's manual).
- Coil operating voltage range percentages (check IEC 62271-100 and the manufacturer's data).
- SF6 density and moisture limits (manufacturer's manual and the project procedure).
- Site withstand test voltages for 220 kV and 500 kV (standard and project procedure).
- Which extra items (grading capacitors, pre-insertion resistors, controlled switching) fit the actual 500 kV breaker.
- The exact interlocking schedule of the substation.
- The standard clause numbers and the current edition of each IEC standard quoted in this book.